STAT 238 - BAYESIAN STATISTICS
HOMEWORK THREE

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(1) Suppose $X_1, \ldots, X_n$ are i.i.d observations from a distribution with finite variance σ^2.

(a) Describe an algorithm based on classical bootstrap to obtain a 95% interval for σ. (**3 points**).

Solution:

(i) For $b = 1, \ldots, B$ (with B large, e.g. $B = 10{,}000$):

(A) Draw a bootstrap sample $X_1^{(b)}, \ldots, X_n^{(b)}$ of size n with replacement from the observed data $X_1, \ldots, X_n$.

(B) Compute the sample standard deviation of the bootstrap sample:

$$\hat{\sigma}^{(b)} = \sqrt{\frac{1}{n-1} \sum_{i=1}^{n} (X_i^{(b)} - \bar{X}^{(b)})^2}.$$

(ii) This produces bootstrap estimates $\hat{\sigma}^{(1)}, \ldots, \hat{\sigma}^{(B)}$.

(iii) Sort these values and take the empirical 2.5% and 97.5% percentiles.

(iv) The interval

$$[\hat{\sigma}_{0.025},\, \hat{\sigma}_{0.975}]$$

is the 95% bootstrap confidence interval for σ.

(b) Describe an algorithm based on the Bayesian bootstrap to obtain a 95% interval for σ. (**3 points**).

Solution:

(i) For $b = 1, \ldots, B$ (with B large, e.g. $B = 10{,}000$):

(A) Draw weights $w^{(b)} = (w_1^{(b)}, \ldots, w_n^{(b)})$ from a Dirichlet$(1, \ldots, 1)$ distribution, so that $w_i^{(b)} \geq 0$ and $\sum_{i=1}^{n} w_i^{(b)} = 1$.

(B) Compute the weighted mean

$$\bar{X}^{(b)} = \sum_{i=1}^{n} w_i^{(b)} X_i.$$

(C) Compute the weighted variance

$$(\sigma_w^2)^{(b)} = \sum_{i=1}^{n} w_i^{(b)} (X_i - \bar{X}^{(b)})^2,$$

and the corresponding weighted standard deviation

$$\hat{\sigma}^{(b)} = \sqrt{(\sigma_w^2)^{(b)}}.$$

(ii) Repeat the above steps to obtain $\hat{\sigma}^{(1)}, \ldots, \hat{\sigma}^{(B)}$.

(iii) The 95% credible interval for σ is given by the empirical 2.5% and 97.5% percentiles of these values.

Date: today.

(iv) The interval

$$[\hat{\sigma}_{0.025},\ \hat{\sigma}_{0.975}]$$

is the 95% bootstrap confidence interval for σ.

(c) Assuming the model $X_1, \ldots, X_n \overset{\text{i.i.d}}{\sim} N(\mu, \sigma^2)$, describe a 95% uncertainty interval (frequentist or Bayesian) for σ. If you want a Bayesian interval, you can directly use problem 3(e) from Homework One (**2 points**).

Solution: Using the frequentist approach or a Bayesian approach with the standard non-informative prior $p(\mu, \sigma^2) \propto 1/\sigma^2$, the posterior/sampling distribution of the variance relies on the scaled chi-square distribution. Specifically,

$$\frac{(n-1)s^2}{\sigma^2} \sim \chi^2_{n-1}$$

where s^2 is the sample variance. To find a 95% interval for σ, we take the 2.5^{th} and 97.5^{th} quantiles of the χ^2_{n-1} distribution (let's call them $q_{0.025}$ and $q_{0.975}$). The 95% interval for σ is then:

$$\left[\sqrt{\frac{(n-1)s^2}{q_{0.975}}}, \sqrt{\frac{(n-1)s^2}{q_{0.025}}}\right]$$

Solution:

Frequentist approach:

Assume $X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} N(\mu, \sigma^2)$. The sample variance

$$s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2$$

satisfies the pivotal relation

$$\frac{(n-1)s^2}{\sigma^2} \sim \chi^2_{n-1}.$$

Let $\chi^2_{n-1,\,0.025}$ and $\chi^2_{n-1,\,0.975}$ denote the 2.5% and 97.5% quantiles of the χ^2_{n-1} distribution. Then

$$P\left(\chi^2_{n-1,\,0.025} \leq \frac{(n-1)s^2}{\sigma^2} \leq \chi^2_{n-1,\,0.975}\right) = 0.95.$$

Solving for σ gives the 95% confidence interval

$$\left[\sqrt{\frac{(n-1)s^2}{\chi^2_{n-1,\,0.975}}},\ \sqrt{\frac{(n-1)s^2}{\chi^2_{n-1,\,0.025}}}\right].$$

Bayesian approach:

Using the non-informative prior

$$p(\mu, \sigma^2) \propto \frac{1}{\sigma^2},$$

the posterior distribution of σ^2 given the data is

$$\frac{(n-1)s^2}{\sigma^2} \mid X_1, \ldots, X_n \sim \chi^2_{n-1}.$$

Thus the posterior credible interval for σ has the same form as the frequentist confidence interval:

$$\left[\sqrt{\frac{(n-1)s^2}{\chi^2_{n-1,\,0.975}}},\ \sqrt{\frac{(n-1)s^2}{\chi^2_{n-1,\,0.025}}}\right].$$

(d) Consider the data in `old_faithful_eruptions.txt` which gives eruption durations for the old faithful geyser in the Yellowstone National Park. Using all three methods above, compute a 95% interval for σ. Compare the resulting intervals: are they similar? which method produces the widest interval? (**6 points**).
Solution: [EMPTY - Requires data file `old_faithful_eruptions.txt`]

(2) Suppose $(X_1, Y_1), \ldots, (X_n, Y_n)$ are i.i.d observations from a distribution with correlation ρ.

(a) Describe an algorithm based on classical bootstrap to obtain a 95% interval for ρ. (**3 points**).
Solution:
(i) Draw a random sample of size n with replacement from the paired data (X_i, Y_i) to form a bootstrap sample.
(ii) Compute the Pearson correlation coefficient $\hat{\rho}^{(b)}$ for this bootstrap sample.
(iii) Repeat steps 1 and 2 a large number of times B.
(iv) Sort the values and take the 2.5^{th} and 97.5^{th} percentiles to form the 95% confidence interval.

(b) Describe an algorithm based on the Bayesian bootstrap to obtain a 95% interval for ρ. (**3 points**).
Solution:
(i) Draw n weights $w = (w_1, \ldots, w_n)$ from a Dirichlet$(1, \ldots, 1)$ distribution.
(ii) Calculate the weighted means: $\bar{X}_w = \sum w_i X_i$ and $\bar{Y}_w = \sum w_i Y_i$.
(iii) Calculate the weighted covariance and variances: $\text{Cov}_w = \sum w_i (X_i - \bar{X}_w)(Y_i - \bar{Y}_w)$, $V_{X,w} = \sum w_i (X_i - \bar{X}_w)^2$, and $V_{Y,w} = \sum w_i (Y_i - \bar{Y}_w)^2$.
(iv) Compute the weighted correlation: $\hat{\rho}^{(b)}_w = \frac{\text{Cov}_w}{\sqrt{V_{X,w} V_{Y,w}}}$.
(v) Repeat steps 1 through 4 B times and extract the 2.5^{th} and 97.5^{th} percentiles.

(c) Consider the data in `PearsonHeightData.txt` which contains heights of adult father-son pairs. Using all three methods above, compute a 95% interval for ρ. Compare the resulting intervals: are they similar? which method produces the widest interval? (**6 points**).
Solution: [EMPTY - Requires data file `PearsonHeightData.txt`]

(3) Consider the kidney cancer dataset from Homework Two and class. Here X_i denotes the total death count due to kidney cancer for county i in the period $1980-89$. Also n_i denotes the average population for county i in the same time period. We want to use the model:

$$X_i \overset{\text{ind}}{\sim} \text{Binomial}(n_i, \theta_i)$$

along with the prior

$$\theta_i \overset{\text{i.i.d}}{\sim} \text{Beta}(a, b).$$

We want to treat a and b as unknown parameters and estimate them by maximizing the marginal likelihood of the data $X_1, \ldots, X_N$. In Lecture 9, we discussed maximization of (1) by taking a grid of values for a and b. The goal of this problem is to solve the same maximization using gradient descent.

(a) Show that the marginal likelihood for a, b is proportional to:

$$\prod_{i=1}^{N} \frac{B(X_i + a, n_i - X_i + b)}{B(a, b)} \tag{1}$$

where N is the total number of counties, and $B(\cdot, \cdot)$ denotes the Beta function. (**3 points**).

Solution: The marginal likelihood is obtained by integrating out θ_i from the joint distribution. For a single county i:

$$p(X_i \mid a, b) = \int_0^1 p(X_i \mid \theta_i) p(\theta_i \mid a, b) d\theta_i$$

$$p(X_i \mid a, b) = \int_0^1 \binom{n_i}{X_i} \theta_i^{X_i} (1 - \theta_i)^{n_i - X_i} \frac{1}{B(a, b)} \theta_i^{a-1} (1 - \theta_i)^{b-1} d\theta_i$$

Factoring out terms independent of θ_i:

$$p(X_i \mid a, b) = \binom{n_i}{X_i} \frac{1}{B(a, b)} \int_0^1 \theta_i^{X_i + a - 1} (1 - \theta_i)^{n_i - X_i + b - 1} d\theta_i$$

The integral is exactly the Beta function $B(X_i + a, n_i - X_i + b)$. Thus:

$$p(X_i \mid a, b) = \binom{n_i}{X_i} \frac{B(X_i + a, n_i - X_i + b)}{B(a, b)}$$

Assuming counties are independent, the total marginal likelihood is the product over all N counties. Dropping the combinatorial constant $\binom{n_i}{X_i}$ which does not depend on a or b, the marginal likelihood is proportional to:

$$\prod_{i=1}^{N} \frac{B(X_i + a, n_i - X_i + b)}{B(a, b)}$$

(b) Write the marginal likelihood (1) using the Gamma function. (**2 points**)

Solution: Recall the relationship between the Beta and Gamma functions: $B(x, y) = \frac{\Gamma(x)\Gamma(y)}{\Gamma(x+y)}$. Substituting this into (1):

$$\prod_{i=1}^{N} \frac{B(X_i + a, n_i - X_i + b)}{B(a, b)} = \prod_{i=1}^{N} \left(\frac{\Gamma(X_i + a)\Gamma(n_i - X_i + b)}{\Gamma(n_i + a + b)} \times \frac{\Gamma(a + b)}{\Gamma(a)\Gamma(b)} \right)$$

(c) Write the gradient (with respect to a, b) of the logarithm of the marginal likelihood in terms of the digamma function. (**2 points**)

Solution: Let $L(a, b)$ be the log marginal likelihood without constants.

$$L(a, b) = \sum_{i=1}^{N} \Big[\log\Gamma(X_i + a) + \log\Gamma(n_i - X_i + b) + \log\Gamma(a + b) - \log\Gamma(n_i + a + b) - \log\Gamma(a) - \log\Gamma(b) \Big]$$

The derivative of the log-Gamma function is the digamma function, $\psi(x) = \frac{d}{dx} \log\Gamma(x)$. Thus, the gradients are:

$$\frac{\partial L}{\partial a} = \sum_{i=1}^{N} \Big[\psi(X_i + a) + \psi(a + b) - \psi(n_i + a + b) - \psi(a) \Big]$$

$$\frac{\partial L}{\partial b} = \sum_{i=1}^{N} \Big[\psi(n_i - X_i + b) + \psi(a + b) - \psi(n_i + a + b) - \psi(b) \Big]$$

(d) Implement a gradient ascent scheme for maximizing the logarithm of the marginal likelihood. You are welcome to use a scheme similar to that used in Lecture 14 to select the right step size. Initialize the optimization with $a = b = 1$. (**5 points**)
Solution: [EMPTY - Requires Kidney Cancer Dataset execution]

(e) Compare the answer obtained by gradient descent with the answer obtained in Lecture 9 via grid based maximization. Are the answers similar. (**2 points**).
Solution: [EMPTY - Requires Kidney Cancer Dataset execution]

(4) Consider the two text files: `spam.txt` and `ham.txt`...
Solution: [EMPTY - Entire Question 4 requires data files `spam.txt` and `ham.txt`]

(5) Consider the text data file `alice.txt`...
Solution: [EMPTY - Entire Question 5 requires data file `alice.txt`]

(6) Consider two binary variables A and B. We observe data...

(a) Model 1 for the joint distribution of (A_i, B_i) uses the decomposition:

$$\mathbb{P}\{A_i = a_i, B_i = b_i\} = \mathbb{P}\{A_i = a_i\}\mathbb{P}\{B_i = b_i \mid A_i = a_i\}.$$

Model 1 is given by:

$$\mathbb{P}\{A_i = 1\}, \mathbb{P}\{B_i = 1 \mid A_i = 1\}, \mathbb{P}\{B_i = 1 \mid A_i = 0\} \overset{\text{i.i.d}}{\sim} \text{uniform}(0, 1).$$

Write down the joint density on $(\theta_{00}, \theta_{01}, \theta_{10}, \theta_{11})$ under Model 1. (**3 points**)
Solution: Let $p_A = \mathbb{P}\{A = 1\}$, $p_{B|1} = \mathbb{P}\{B = 1|A = 1\}$, and $p_{B|0} = \mathbb{P}\{B = 1|A = 0\}$. The transformation mapping is:

$$\theta_{11} = p_A p_{B|1}, \quad \theta_{10} = p_A(1 - p_{B|1})$$
$$\theta_{01} = (1 - p_A)p_{B|0}, \quad \theta_{00} = (1 - p_A)(1 - p_{B|0})$$

Since $p_A, p_{B|1}, p_{B|0} \sim U(0, 1)$, their joint density is 1. We apply the change of variables formula. Notice that $p_A = \theta_{11}+\theta_{10}$, $p_{B|1} = \frac{\theta_{11}}{\theta_{11}+\theta_{10}}$, and $p_{B|0} = \frac{\theta_{01}}{\theta_{01}+\theta_{00}}$. The determinant of the Jacobian of the inverse transformation is given by $\frac{1}{p_A(1-p_A)} = \frac{1}{(\theta_{11}+\theta_{10})(\theta_{01}+\theta_{00})}$. Therefore, the joint density on the simplex under Model 1 is:

$$f_1(\theta_{00}, \theta_{01}, \theta_{10}, \theta_{11}) = \frac{1}{(\theta_{11} + \theta_{10})(\theta_{01} + \theta_{00})}$$

(b) Model 2 for the joint distribution of (A_i, B_i) uses the decomposition:

$$\mathbb{P}\{A_i = a_i, B_i = b_i\} = \mathbb{P}\{B_i = b_i\}\mathbb{P}\{A_i = a_i \mid B_i = b_i\}.$$

Model 2 is given by:

$$\mathbb{P}\{B_i = 1\}, \mathbb{P}\{A_i = 1 \mid B_i = 1\}, \mathbb{P}\{A_i = 1 \mid B_i = 0\} \overset{\text{i.i.d}}{\sim} \text{uniform}(0, 1).$$

Write down the joint density on $(\theta_{00}, \theta_{01}, \theta_{10}, \theta_{11})$ under Model 2. (**3 points**)
Solution: By symmetric logic, switching the roles of variables A and B, the determinant of the Jacobian will feature the marginals of B. $P(B = 1) = \theta_{11} + \theta_{01}$ and $P(B = 0) = \theta_{10} + \theta_{00}$. Thus, the joint density on the simplex under Model 2 is:

$$f_2(\theta_{00}, \theta_{01}, \theta_{10}, \theta_{11}) = \frac{1}{(\theta_{11} + \theta_{01})(\theta_{10} + \theta_{00})}$$

(c) Under each model, calculate the marginal density of $\mathbb{P}\{A_i = 1\}$. For Model 2, is this marginal density uniform on $(0, 1)$? (**3 points**)
Solution: Under Model 1, $\mathbb{P}\{A_i = 1\} = p_A$, which is explicitly defined as uniform$(0, 1)$ in the problem description.
Under Model 2, $\mathbb{P}\{A_i = 1\} = \mathbb{P}\{B = 1\}\mathbb{P}\{A = 1|B = 1\} + \mathbb{P}\{B = 0\}\mathbb{P}\{A = 1|B = 0\} = p_B p_{A|1} + (1 - p_B)p_{A|0}$. Since this is a sum of products of independent uniform

random variables, its resulting distribution is strongly peaked at 0.5 and goes to zero at the boundaries 0 and 1. Therefore, no, the marginal density for $\mathbb{P}\{A_i = 1\}$ under Model 2 is **not** uniform on $(0, 1)$.

(d) Given the observed data $(A_i, B_i), i = 1, \ldots, n$, let $N_{00}, N_{01}, N_{10}, N_{11}$ denote the observed counts for the four cases $(0, 0), (0, 1), (1, 0), (1, 1)$. Write down the posterior density of $(\theta_{00}, \theta_{01}, \theta_{10}, \theta_{11})$ under each model (Model 1 and Model 2). (**4 points**).
Solution: The likelihood for the data is proportional to $L(\theta) \propto \theta_{00}^{N_{00}}\theta_{01}^{N_{01}}\theta_{10}^{N_{10}}\theta_{11}^{N_{11}}$. Multiplying by the respective prior densities derived in parts (a) and (b):
Model 1 Posterior:

$$p(\theta \mid \text{data}, M_1) \propto \frac{\theta_{00}^{N_{00}}\theta_{01}^{N_{01}}\theta_{10}^{N_{10}}\theta_{11}^{N_{11}}}{(\theta_{11} + \theta_{10})(\theta_{01} + \theta_{00})}$$

Model 2 Posterior:

$$p(\theta \mid \text{data}, M_2) \propto \frac{\theta_{00}^{N_{00}}\theta_{01}^{N_{01}}\theta_{10}^{N_{10}}\theta_{11}^{N_{11}}}{(\theta_{11} + \theta_{01})(\theta_{10} + \theta_{00})}$$

(e) Calculate the posterior distribution of I given the data $(A_1, B_1), \ldots, (A_n, B_n)$. This can be interpreted as the posterior probabilities associated with the two models given the observed data. (**4 points**)
Solution: Using Bayes' Rule, $P(I = i \mid \text{data}) \propto P(\text{data} \mid I = i)P(I = i)$. Since $P(I = 1) = P(I = 2) = 0.5$, the posterior ratio depends directly on the ratio of the marginal likelihoods $P(\text{data} \mid I = i)$.
For Model 1, the likelihood perfectly factors over the independent uniforms $p_A, p_{B|1}, p_{B|0}$:

$$P(\text{data} \mid M_1) = \int p_A^{N_{1\cdot}}(1 - p_A)^{N_{0\cdot}} p_{B|1}^{N_{11}}(1 - p_{B|1})^{N_{10}} p_{B|0}^{N_{01}}(1 - p_{B|0})^{N_{00}} dp_A dp_{B|1} dp_{B|0}$$

$$= B(N_{1\cdot} + 1, N_{0\cdot} + 1)B(N_{11} + 1, N_{10} + 1)B(N_{01} + 1, N_{00} + 1)$$

where $N_{1\cdot} = N_{11} + N_{10}$ and $N_{0\cdot} = N_{01} + N_{00}$.
For Model 2, factoring analogously over B:

$$P(\text{data} \mid M_2) = B(N_{\cdot 1} + 1, N_{\cdot 0} + 1)B(N_{11} + 1, N_{01} + 1)B(N_{10} + 1, N_{00} + 1)$$

The posterior probabilities are thus:

$$P(I = 1 \mid \text{data}) = \frac{P(\text{data} \mid M_1)}{P(\text{data} \mid M_1) + P(\text{data} \mid M_2)}$$

$$P(I = 2 \mid \text{data}) = \frac{P(\text{data} \mid M_2)}{P(\text{data} \mid M_1) + P(\text{data} \mid M_2)}$$

(f) Calculate the posterior distribution of I given the data, and show that the posterior probability of Model 1 (i.e., of $I = 1$) is about 3.8 times the posterior probability of Model 2 (i.e., of $I = 2$). (**3 points**).
Solution: From Table 1, we have: $N_{00} = 760$, $N_{01} = 190$, $N_{10} = 5$, $N_{11} = 45$. Row totals: $N_{0\cdot} = 950$, $N_{1\cdot} = 50$. Column totals: $N_{\cdot 0} = 765$, $N_{\cdot 1} = 235$.
We compute the Bayes Factor $K = \frac{P(\text{data}|M_1)}{P(\text{data}|M_2)}$.

$$P(\text{data} \mid M_1) = B(51, 951)B(46, 6)B(191, 761)$$

$$P(\text{data} \mid M_2) = B(236, 766)B(46, 191)B(6, 761)$$

Using $B(x,y) = \frac{\Gamma(x)\Gamma(y)}{\Gamma(x+y)}$:

$$K = \frac{\frac{\Gamma(51)\Gamma(951)}{\Gamma(1002)} \frac{\Gamma(46)\Gamma(6)}{\Gamma(52)} \frac{\Gamma(191)\Gamma(761)}{\Gamma(952)}}{\frac{\Gamma(236)\Gamma(766)}{\Gamma(1002)} \frac{\Gamma(46)\Gamma(191)}{\Gamma(237)} \frac{\Gamma(6)\Gamma(761)}{\Gamma(767)}}$$

Notice that the terms $\Gamma(1002)$, $\Gamma(46)$, $\Gamma(6)$, $\Gamma(191)$, and $\Gamma(761)$ cancel neatly from the numerator and denominator:

$$K = \frac{\Gamma(51)\Gamma(951)}{\Gamma(52)\Gamma(952)} \times \frac{\Gamma(237)\Gamma(767)}{\Gamma(236)\Gamma(766)}$$

Applying the property $\Gamma(z+1) = z\Gamma(z)$:

$$K = \left(\frac{1}{51 \times 951}\right) \times (236 \times 766) = \frac{180776}{48501} \approx 3.727$$

Thus, the posterior probability of Model 1 is approximately 3.73 (or about 3.8) times that of Model 2.

(g) [**Bonus question**] ... Discuss your thoughts on this interpretation of the posterior comparison, particularly in relation to whether such calculations genuinely provide evidence about causal direction. **(3 points)**

Solution: While the Bayes Factor strongly favors Model 1, this does **not** provide genuine evidence of causal direction ($A \to B$). Instead, it simply reflects that under uniform priors, Model 1's parameterization is a more compact/plausible fit to the specific marginals of this contingency table than Model 2's. Both models are mathematically equivalent at describing the joint distribution Space, but their priors induce distinct distributions over the space of joint probabilities (as proved in parts a-c). We are effectively comparing which set of prior assumptions better aligns with the data, rather than uncovering an inherent physical or causal asymmetry. Causal inference typically requires domain knowledge, intervention (experimental data), or structural causal assumptions beyond just observational correlation mappings.

(7) Consider the multiple regression model...

(a) Prove that

$$\beta \mid \text{data}, \sigma \sim N_{m+1}\left(\hat{\beta}, \sigma^2 \left(X^T X\right)^{-1}\right). \tag{2}$$

Solution: The likelihood of the data is given by $p(y \mid \beta, \sigma) \propto (\sigma^2)^{-n/2} \exp\left(-\frac{1}{2\sigma^2}(y - X\beta)^T(y - X$
Using the identity $(y - X\beta)^T(y - X\beta) = (y - X\hat{\beta})^T(y - X\hat{\beta}) + (\beta - \hat{\beta})^T X^T X(\beta - \hat{\beta}) = S(\hat{\beta}) + (\beta - \hat{\beta})^T X^T X(\beta - \hat{\beta})$, where $\hat{\beta} = (X^T X)^{-1} X^T y$. Given the flat prior on β, $p(\beta \mid \sigma) \propto 1$ (as $C \to \infty$), the conditional posterior is proportional to the likelihood viewed as a function of β:

$$p(\beta \mid \text{data}, \sigma) \propto \exp\left(-\frac{1}{2\sigma^2}(\beta - \hat{\beta})^T X^T X(\beta - \hat{\beta})\right)$$

This is precisely the kernel of a multivariate Normal distribution with mean $\hat{\beta}$ and covariance matrix $\sigma^2(X^T X)^{-1}$.

(b) Prove that the posterior density of σ (i.e., the density of $\sigma \mid$ data) is proportional to

$$\sigma^{-n+m} \exp\left(-\frac{S(\hat{\beta})}{2\sigma^2}\right) I\{\sigma > 0\}, \tag{3}$$

where $S(\hat{\beta})$ is the Residual Sum of Squares. (**4 points**)

Solution: The prior on σ implied by $\log\sigma \sim \text{Unif}$ is $p(\sigma) \propto 1/\sigma$. The joint posterior $p(\beta, \sigma \mid y)$ is proportional to likelihood $\times$ prior:

$$p(\beta, \sigma \mid y) \propto \sigma^{-n-1} \exp\left(-\frac{S(\hat{\beta})}{2\sigma^2}\right) \exp\left(-\frac{1}{2\sigma^2}(\beta - \hat{\beta})^T X^T X(\beta - \hat{\beta})\right)$$

To find $p(\sigma \mid y)$, we integrate out the $m+1$ dimensional parameter β:

$$p(\sigma \mid y) = \int p(\beta, \sigma \mid y) d\beta \propto \sigma^{-n-1} \exp\left(-\frac{S(\hat{\beta})}{2\sigma^2}\right) \int \exp\left(-\frac{1}{2\sigma^2}(\beta - \hat{\beta})^T X^T X(\beta - \hat{\beta})\right) d\beta$$

The integral is the normalizing constant of a multivariate Normal, which evaluates to $(2\pi)^{(m+1)/2}|\sigma^2(X^TX)^{-1}|^{1/2} \propto (\sigma^2)^{(m+1)/2} = \sigma^{m+1}$. Thus:

$$p(\sigma \mid y) \propto \sigma^{-n-1}\sigma^{m+1} \exp\left(-\frac{S(\hat{\beta})}{2\sigma^2}\right) = \sigma^{-n+m} \exp\left(-\frac{S(\hat{\beta})}{2\sigma^2}\right)$$

for $\sigma > 0$.

(c) Use the results of the previous two parts... to deduce (**??**). (**4 points**)

Solution: We integrate the joint posterior $p(\beta, \sigma \mid y) = p(\beta \mid y, \sigma)p(\sigma \mid y)$ over σ:

$$f_{\beta|\text{data}}(\beta) \propto \int_0^\infty \sigma^{-(m+1)} \exp\left(-\frac{Q(\beta)}{2\sigma^2}\right) \sigma^{-n+m} \exp\left(-\frac{S(\hat{\beta})}{2\sigma^2}\right) d\sigma$$

where $Q(\beta) = (\beta - \hat{\beta})^T X^T X(\beta - \hat{\beta})$. Combining exponents:

$$\propto \int_0^\infty \sigma^{-(n+1)} \exp\left(-\frac{S(\hat{\beta}) + Q(\beta)}{2\sigma^2}\right) d\sigma$$

Using the change of variables $u = 1/\sigma^2$, this is a Gamma integral evaluating to $(S(\hat{\beta}) + Q(\beta))^{-n/2}$. We rewrite this as:

$$\propto \left(1 + \frac{1}{n-m-1}(\beta - \hat{\beta})^T \frac{X^TX}{\hat{\sigma}^2}(\beta - \hat{\beta})\right)^{-\frac{(n-m-1)+(m+1)}{2}}$$

where $\hat{\sigma}^2 = \frac{S(\hat{\beta})}{n-m-1}$. This is the kernel of a multivariate t-distribution with $n-m-1$ degrees of freedom, location $\hat{\beta}$, and scale matrix $\hat{\sigma}^2(X^TX)^{-1}$.

(d) Use the result in part (b) to show that (**3 points**)

$$\frac{S(\hat{\beta})}{\sigma^2} \mid \text{data} \sim \chi^2_{n-m-1}.$$

Solution: Let $Q = \frac{S(\hat{\beta})}{\sigma^2}$. Then $\sigma = \sqrt{\frac{S(\hat{\beta})}{Q}}$ and $d\sigma = -\frac{1}{2}\sqrt{S(\hat{\beta})}Q^{-3/2}dQ$. Using the change of variables formula on the density from part (b):

$$f_Q(Q) \propto \left(\sqrt{\frac{S}{Q}}\right)^{-n+m} \exp\left(-\frac{Q}{2}\right) Q^{-3/2} = Q^{\frac{n-m-1}{2}-1} \exp(-Q/2)$$

This is exactly the kernel of a Chi-squared distribution with $n-m-1$ degrees of freedom.

(e) Use the result in the previous part to argue that the usual estimator $\hat{\sigma}$ is a reasonable point estimate of σ. (**2 points**).
Solution: The mean of a χ^2_k distribution is k. Thus, the posterior expectation $E\left[\frac{S(\hat{\beta})}{\sigma^2} \mid \text{data}\right] = n - m - 1$. Inverting this relationship intuitively gives an estimator for σ^2 as $\frac{S(\hat{\beta})}{n-m-1}$, which matches the classic unbiased frequentist estimator $\hat{\sigma}^2$. Its square root $\hat{\sigma}$ serves as a very reasonable point estimate for σ.

(8) This problem discusses predictions in linear regression...

(a) Suppose we are interested in the parameter $a^T\beta$ for some fixed vector a. Using (2), prove that... (**3 points**)
Solution: Since $\beta \mid \text{data}, \sigma \sim N_{m+1}(\hat{\beta}, \sigma^2(X^TX)^{-1})$, any linear combination $a^T\beta$ is univariate Normal. The mean is simply $a^T\mathbb{E}[\beta] = a^T\hat{\beta}$. The variance is $a^T\text{Var}(\beta)a = a^T(\sigma^2(X^TX)^{-1})a = \sigma^2 a^T(X^TX)^{-1}a$. Therefore,

$$a^T\beta \mid \text{data}, \sigma \sim N\left(a^T\hat{\beta}, \sigma^2 a^T\left(X^TX\right)^{-1}a\right)$$

(b) Using (**??**), prove that (**3 points**)
Solution: Equation (**??**) establishes that the marginal posterior of β is a multivariate t-distribution $t_{n-m-1,m+1}(\hat{\beta}, \hat{\sigma}^2(X^TX)^{-1})$. A linear combination of a multivariate t-distributed random vector results in a univariate t-distribution with the same degrees of freedom. The location translates to $a^T\hat{\beta}$ and the scale is geometrically transformed by the vector a. Thus:

$$a^T\beta \mid \text{data} \sim t_{n-m-1}\left(a^T\hat{\beta}, \hat{\sigma}^2 a^T(X^TX)^{-1}a\right)$$

(c) Suppose we are interested in predicting the value Y_{n+1}... prove that (**4 points**)
Solution: The new observation is $Y_{n+1} = \tilde{x}^T_{n+1}\beta + \epsilon_{n+1}$. Given the data and σ, β and ϵ_{n+1} are independent. We know the distributions: $\tilde{x}^T_{n+1}\beta \mid \text{data}, \sigma \sim N(\tilde{x}^T_{n+1}\hat{\beta}, \sigma^2\tilde{x}^T_{n+1}(X^TX)^{-1}\tilde{x}_{n+}$ and $\epsilon_{n+1} \sim N(0, \sigma^2)$. The sum of two independent normal distributions is normal. We sum the means and the variances:

$$\mathbb{E}[Y_{n+1} \mid \text{data}, \sigma] = \tilde{x}^T_{n+1}\hat{\beta} + 0$$

$$\text{Var}(Y_{n+1} \mid \text{data}, \sigma) = \sigma^2\tilde{x}^T_{n+1}(X^TX)^{-1}\tilde{x}_{n+1} + \sigma^2 = \sigma^2\left(1 + \tilde{x}^T_{n+1}(X^TX)^{-1}\tilde{x}_{n+1}\right)$$

Thus, $Y_{n+1} \mid \text{data}, \sigma \sim N\left(\tilde{x}^T_{n+1}\hat{\beta}, \sigma^2 + \sigma^2\tilde{x}^T_{n+1}\left(X^TX\right)^{-1}\tilde{x}_{n+1}\right)$.

(d) Using the result from the previous part... prove that
Solution: We integrate the conditional density of Y_{n+1} given σ over the marginal posterior density of σ:

$$f_{Y_{n+1}|\text{data}}(y) = \int_0^\infty N(y \mid \tilde{x}^T_{n+1}\hat{\beta}, \sigma^2 c) \times p(\sigma \mid \text{data})d\sigma$$

where $c = 1 + \tilde{x}^T_{n+1}(X^TX)^{-1}\tilde{x}_{n+1}$. This integral mimics exactly the one in Question 7(c), just mapped to 1 dimension with a modified scale parameter scaling σ^2 by c. The marginalization over σ using an Inverse-Gamma type density for σ^2 yields the Student-t distribution with $n - m - 1$ degrees of freedom. Substituting the expectation and scale parameters:

$$Y_{n+1} \mid \text{data} \sim t_{n-m-1}\left(\tilde{x}^T_{n+1}\hat{\beta}, \hat{\sigma}^2 + \hat{\sigma}^2\tilde{x}^T_{n+1}\left(X^TX\right)^{-1}\tilde{x}_{n+1}\right)$$

References

[1] MacKay, D. J. C., (2003) *Information theory, inference, and learning algorithms*. Cambridge University Press, 2003.